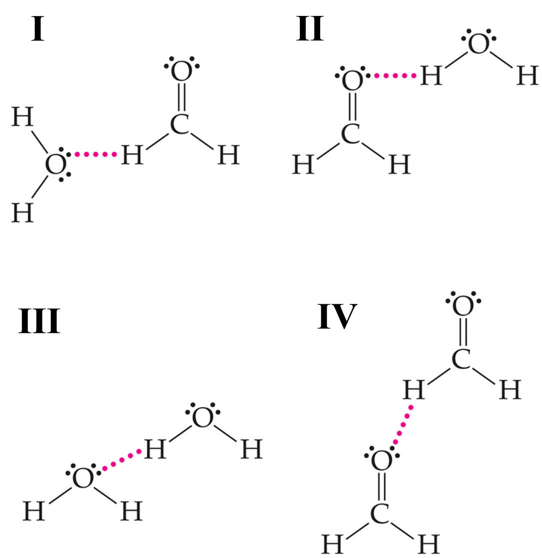
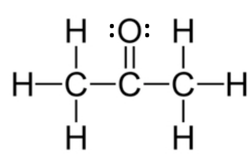
1. Which image correctly depicts the hydrogen bonding interactions that can occur in an aqueous solution of formaldehyde (CH_2O)?



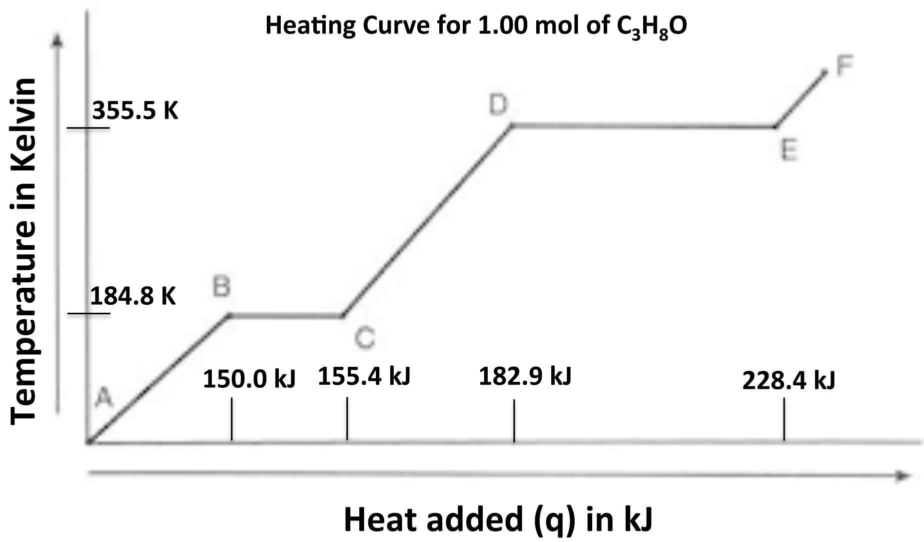
A) I only
B) I, II, and IV
C) II only
D) II and III
E) All of I, II, III, and IV

2. Arrange the molecules in order of *increasing* vapor pressure at 1 atm:
CH_3COOH, C_2H_6, C_4H_{10}, C_3H_6O (see image below)



A) $CH_3COOH < C_2H_6 < C_4H_{10} < C_3H_6O$
B) $C_2H_6 < C_4H_{10} < C_3H_6O < CH_3COOH$
C) $CH_3COOH < C_3H_6O < C_4H_{10} < C_2H_6$
D) $C_2H_6 < C_4H_{10} < CH_3COOH < C_3H_6O$
E) $C_4H_{10} < C_2H_6 < C_3H_6O < CH_3COOH$

3. Which statement is true regarding the heating curve below? Note that C_3H_8O has only one solid, one liquid, and one gas phase under the given conditions.



A) The inverse of the slope of line segment A—B represents the molar heat capacity of the gas.
B) The temperature associated with point D represents the melting point for the substance.
C) The inverse of the slope of line segment C—D represents the molar heat capacity for the liquid.
D) Line segment E—F represents the heat of vaporization for the solid.
E) The boiling point for the substance is 184.8 K.

4. How much heat (in kJ) is required to convert 1.60 moles of benzene (C_6H_6) from a liquid at 296 K to gas at 353.3 K?

Melting point = 278.6 K
Boiling point = 353.3 K
Molar heat capacity of the solid = 118 J/mol•K
Molar heat capacity of the liquid = 135 J/mol•K
Molar heat capacity of the gas = 104 J/mol•K
ΔH^{o}_{fusion} = 9.9 kJ/mol
$\Delta H^{o}_{vaporization}$ = 30.7 kJ/mol

A) 12.4 kJ
B) 18.8 kJ
C) 49.12 kJ
D) 61.5 kJ
E) 1360 kJ

5. Lead (Pb) crystallizes in a face-centered cubic unit cell. What is the mass of the cubic unit cell?

   A) 6.02×10^{-23} g
   B) 3.44×10^{-22} g
   C) 6.88×10^{-22} g
   D) 1.38×10^{-21} g
   E) 4.82×10^{-21} g

6. Which statement is true regarding a semi-conductor made of germanium (Ge) doped with arsenic (As)?

   A) Electrons are removed creating an "acceptor level" that is close in energy to the conduction band. This increases conductivity.
   B) Electrons are added in a "donor level" that is close in energy to the conduction band. This increases conductivity.
   C) Electrons are added in an "acceptor level" that is close in energy to the valence band. This decreases conductivity.
   D) Electrons are added in a "donor level" that is close in energy to the valence band. This increases conductivity.
   E) Electrons are removed from the "acceptor level" that is close in energy to the conduction band. This decreases conductivity.

7. Which statement best explains why hexane, C_6H_{14}, is not soluble in water?

   A) The interactions between hexane molecules and water molecules are weaker than the interactions among hexane molecules and the interactions among water molecules.
   B) The interactions between hexane molecules and water molecules are stronger than the interactions among hexane molecules and the interactions among water molecules.
   C) The interactions between hexane molecules and water molecules are equal in strength to the interactions among hexane molecules and the interactions among water molecules.
   D) The interactions among hexane molecules and the interactions among water molecules are weaker than the interactions between hexane molecules and water molecules.
   E) None of answer options A—D. Hexane is soluble in water.

8. The rate law for the reaction 3 A(g) + 2B(g) → C (g) is experimentally determined to be rate = $k[A]^2[B]$. The diagrams below represent different initial mixtures of A and B, and all diagrams can be assumed to represent equal volumes. By what factor does the rate increase from image I to image II?

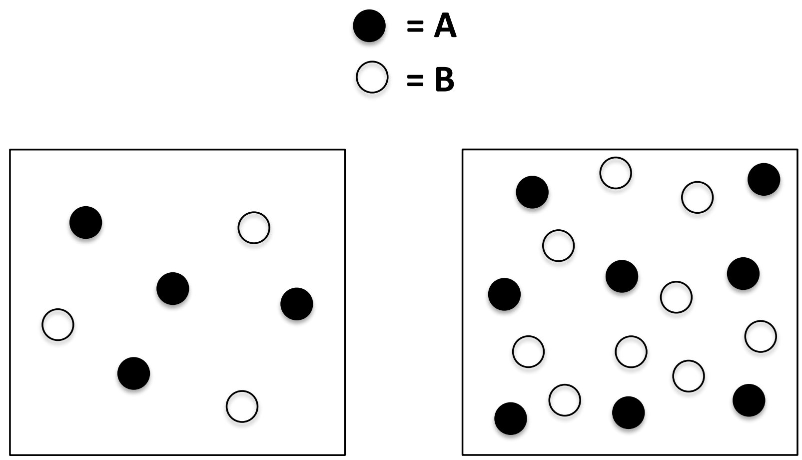


Image I **Image II**

A) 4
B) 6
C) 12
D) 24
E) 72

9. Determine the rate law for the reaction represented by the experimental data:

$$2\,A(aq) + B(aq) \rightarrow C$$

$[A]_{initial}$ (M)	$[B]_{initial}$ (M)	Initial rate (M/s)
0.10	0.10	2.0×10^{-2}
0.20	0.10	8.0×10^{-2}
0.30	0.10	1.8×10^{-1}
0.20	0.20	8.0×10^{-2}

A) Rate = $k[A]$
B) Rate = $k[A]^2$
C) Rate = $k[B]$
D) Rate = $k[B]^2$
E) Rate = $k[A][B]$

10. The decomposition of $SO_2Cl_2(g)$ to form $SO_2(g)$ and $Cl_2(g)$ is first order with $k = 2.90 \times 10^{-4}\ s^{-1}$ at a given temperature. If the concentration of SO_2Cl_2 is 0.0185 *M* after 775 s, then what was the initial concentration of SO_2Cl_2?

A) 0.0148 *M*
B) 0.0232 *M*
C) 0.0333 *M*
D) 0.205 *M*
E) 0.243 *M*

11. The experimentally determined rate law for the reaction AB + C → A + BC is: rate = $k[AB]^2$. Which statement is true regarding the proposed mechanism:

1) AB + AB → AB_2 + A
2) AB_2 + C → AB + BC

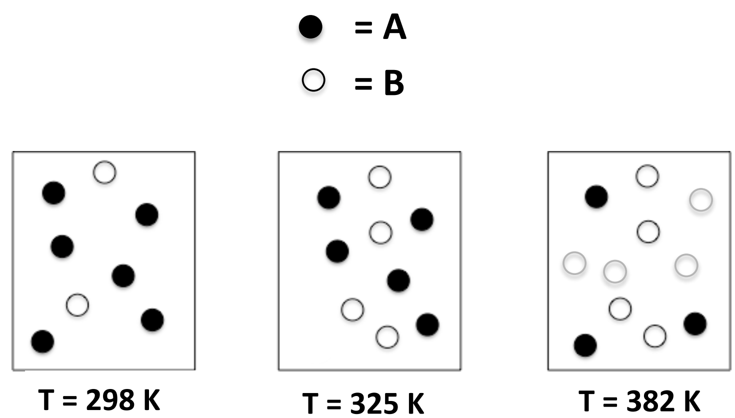
A) Step 1 is the slow step.
B) Step 2 is the slow step.
C) Both steps have the same rate.
D) The relative rates of the two steps cannot be determined.
E) This mechanism cannot be correlated to the overall reaction.

12. Kinetic data is collected, and ln *k* vs. 1/T is graphed for reactions 1 and 2. Which statement is true given the equations for the two graphs?

Reaction 1: y = -22337x + 25.215
Reaction 2: y = -12459x + 31.603

A) The activation energies for both reactions are negative.
B) The frequency factor (pre-exponential factor) for reaction 1 is greater than that for reaction 2.
C) The energy of the transition state for reaction 1 is greater than that for reaction 2.
D) The activation energy for reaction 2 is greater than that for reaction 1.
E) Given similar concentrations of reactants, reaction 1 is faster than reaction 2.

13. Consider the reaction $A(g) \rightleftharpoons 2B(g)$. The images below represent equilibrium mixtures for an endothermic reaction at three different temperatures. All diagrams can be assumed to represent equal volumes. Which statement is true regarding K_c vales for the reaction?



A) All of the reaction mixtures have the same value of K_c.
B) The value of K_c at 298 K is greater than at 325 K but less than at 382 K.
C) The value of K_c is greater at 325 K than at 382 K.
D) The value of K_c is greatest at 382 K.
E) More than one of these statements is true.

14. The reaction between $I_2(g)$ and $Cl_2(g)$ produces ICl(g). At 298 K:

$$I_2(g) + Cl_2(g) \rightleftharpoons 2\,ICl(g) \qquad K_p = 81.9$$

What is the partial pressure of ICl(g) at equilibrium if a reaction mixture initially contains $I_2(g)$ and $Cl_2(g)$ both at partial pressures of 0.15 atm?

A) 0.100 atm
B) 0.123 atm
C) 0.200 atm
D) 0.246 atm
E) 1.75 atm

15. Which "stresses" to the system at equilibrium cause the reaction in which $C(s)$ is converted to $CH_4(g)$ to produce more products?

$$C(s) + 2\,H_2(g) \rightleftharpoons CH_4(g) \qquad \Delta H < 0$$

A) Increase in temperature while holding pressure and volume constant
B) Decrease in pressure while holding temperature constant
C) Addition of $Ar(g)$
D) Addition of $C(s)$
E) Increasing the partial pressure of $H_2(g)$

16. A 0.0321 mole sample of Ag_2SO_4 is dissolved in 1.0 L of water. Which statement regarding this sample is true? All answer options refer to processes carried out at 298 K.

$$Ag_2SO_4(s) \rightleftharpoons 2Ag^+(aq) + SO_4^{2-}(aq) \qquad K_c = 1.1 \times 10^{-5} \text{ at } 298\text{ K}$$

A) More Ag_2SO_4 can be dissolved in the solution because $Q_c < K_c$.
B) No more Ag_2SO_4 can be dissolved in the solution because $Q_c < K_c$.
C) More Ag_2SO_4 can be dissolved in the solution because $Q_c > K_c$.
D) No more Ag_2SO_4 can be dissolved in the solution because $Q_c > K_c$.
E) There is not enough information to determine if more Ag_2SO_4 will dissolve.

17. What is the conjugate acid of HPO_4^{2-}?

A) PO_4^{3-}
B) H_3PO_4
C) $H_2PO_4^-$
D) HPO_5^{2-}
E) H_3O^+

18. Which answer choice is the strongest base? Use the information in the table provided.

Formula of Base	K_b
$(CH_3)_2NH$	5.4×10^{-4}
N_2H_4	8.9×10^{-7}
NH_3	1.8×10^{-5}
$C_6H_5NH_2$	4.27×10^{-10}
$(CH_3)_3N$	6.31×10^{-5}

A) NH_3
B) N_2H_4
C) $(CH_3)_2NH$
D) $C_6H_5NH_2$
E) $(CH_3)_3N$

19. The pH of a solution of KOH is 13.40. What is the concentration of the KOH?

A) 4.0×10^{-14} *M*
B) 1.5×10^{-6} *M*
C) 0.25 *M*
D) 2.6 *M*
E) The K_b of KOH is required to solve the problem.

20. What is the pH of a 0.75 *M* solution of hypochlorous acid, HClO, if the K_a for HClO is 2.9×10^{-8}?

A) 0.13
B) 0.17
C) 1.25
D) 3.83
E) 6.88

21. Which form of the exam do you have?

A)
B)